WORKED SOLUTION

STEP I 2000, Question 4

Pure • Integration

STEP I 2000, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s1-q4

The key idea

We can't integrate $1/(x^2+1)^4$ easily by elementary means, so the question splits it into two pieces: an exact derivative, which integrates for free, and a leftover term that is small. Part (ii) builds the split and part (i) supplies the bound on the leftover. To bound an integral, we usually bound the integrand and multiply by the length of the interval.

Solution

Part (i)

Write $f(x) = \dfrac{x^6}{(x^2+1)^4}$. Since f depends only on x^2, put $t = x^2$, so that t runs over $[0,1]$ as x does and $f = g(t)$ with

$$g(t) = \frac{t^3}{(1+t)^4}.$$

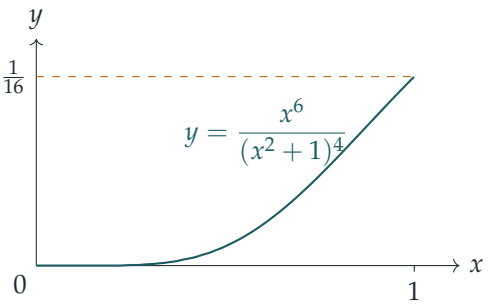
By the quotient rule,

$$g'(t) = \frac{3t^2(1+t)^4 - 4t^3(1+t)^3}{(1+t)^8} = \frac{t^2(3(1+t)-4t)}{(1+t)^5} = \frac{t^2(3-t)}{(1+t)^5}.$$

For $0 < t \leq 1$ every factor is positive (in particular $3-t \geq 2$), so $g'(t) > 0$ and g is strictly increasing on $[0,1]$. Its largest value is therefore at $t = 1$, that is $x = 1$:

$$f(1) = \frac{1}{2^4} = \frac{1}{16},$$

as required. We also have $f(x) \geq 0$ throughout, with $f(0) = 0$ the smallest value.



Part (ii)

Differentiate the bracket with the quotient rule. With $N = Ax^5 + Bx^3 + Cx$ and denominator $(x^2+1)^3$, whose derivative is $6x(x^2+1)^2$, one factor of $(x^2+1)^2$ cancels:

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{N}{(x^2+1)^3}\right) = \frac{(5Ax^4+3Bx^2+C)(x^2+1) - 6x(Ax^5+Bx^3+Cx)}{(x^2+1)^4}.$$

Expanding the numerator and collecting powers of x:

$$\begin{aligned} &5Ax^6 + 5Ax^4 + 3Bx^4 + 3Bx^2 + Cx^2 + C - 6Ax^6 - 6Bx^4 - 6Cx^2 \\ &\quad = -Ax^6 + (5A-3B)x^4 + (3B-5C)x^2 + C. \end{aligned}$$

Everything now has denominator $(x^2+1)^4$, so the identity holds for all x exactly when the numerators agree as polynomials:

$$(D-A)x^6 + (5A-3B)x^4 + (3B-5C)x^2 + C = 1.$$

Comparing coefficients from the bottom up: $C = 1$; then $3B = 5C = 5$; then $5A = 3B = 5$; then $D = A$. So

$$A = 1, \qquad B = \tfrac{5}{3}, \qquad C = 1, \qquad D = 1.$$

Part (iii)

Integrate the identity from part (ii) over $[0,1]$. The first term on the right is an exact derivative, so

$$\begin{aligned} \int_0^1 \frac{\mathrm{d}x}{(x^2+1)^4} &= \left[\frac{x^5 + \frac{5}{3}x^3 + x}{(x^2+1)^3}\right]_0^1 + \int_0^1 \frac{x^6}{(x^2+1)^4}\,\mathrm{d}x \\ &= \frac{1+\frac{5}{3}+1}{8} - 0 + J = \frac{11}{24} + J, \end{aligned}$$

where $J = \displaystyle\int_0^1 \frac{x^6}{(x^2+1)^4}\,\mathrm{d}x$.

Ask yourself How do we bound J without working it out?

By part (i), the integrand of J satisfies $0 \le f(x) \le \frac{1}{16}$ for every x in $[0,1]$. Integrating these inequalities over an interval of length 1 gives

$$0 \le J \le \frac{1}{16} \times 1 = \frac{1}{16}.$$

Adding $\frac{11}{24}$ throughout,

$$\frac{11}{24} \le \int_0^1 \frac{\mathrm{d}x}{(x^2+1)^4} \le \frac{11}{24} + \frac{1}{16},$$

which is the required result. (Both inequalities are in fact strict, since f is continuous and equals neither bound on the whole interval, but the question only asks for $\le$.)

Checks

- Put $x = 0$ in the identity of part (ii): the derivative term is $C = 1$ and the D term vanishes, giving $1 = 1$. At $x = 1$ the derivative's numerator is $-1 + 0 + 0 + 1 = 0$, so the identity reads $\frac{1}{16} = \frac{D}{16}$, consistent with $D = 1$.
- Numerically, Simpson's rule gives $\displaystyle\int_0^1 \frac{\mathrm{d}x}{(x^2+1)^4} \approx 0.47460$, comfortably between $\frac{11}{24} \approx 0.45833$ and $\frac{11}{24} + \frac{1}{16} \approx 0.52083$. The leftover $J \approx 0.01627$ is much less than $\frac{1}{16}$, since f is tiny except near $x = 1$.

Common slips

- Forgetting to cancel the common factor $(x^2+1)^2$ in the quotient rule, which leaves a denominator of $(x^2+1)^6$ and makes the coefficient comparison a mess.
- In part (i), only checking that $x = 1$ is a stationary point or an endpoint: you need to show f is increasing (or compare all candidates) to know the endpoint value is the largest.

Beyond STEP

The substitution $x = \tan\theta$ turns the integral into $\int_0^{\pi/4} \cos^6\theta \, \mathrm{d}\theta$, which can be done exactly with a reduction formula: the value is $\frac{11}{48} + \frac{5\pi}{64}$, matching our numerical estimate. Splitting an integrand into "exact derivative plus something small" is the idea behind reduction formulae, and bounding integrals by bounding integrands is the basic estimate of real analysis. Bartle and Sherbert, *Introduction to Real Analysis*, develops these estimates carefully.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.